Functional Requirements

FR1: User Query Input

The system shall allow users to enter **natural language questions** through a web interface, including questions on dynamic or time-sensitive topics.

FR2: Query Preprocessing

The system shall **clean, normalize, and classify** the query (e.g. time-sensitive or static) before further processing.

FR3: Knowledge Freshness Estimation

The system shall estimate the **likelihood that the LLM's internal knowledge is outdated** for a given query.

FR4: Freshness Score Generation

The system shall output a **numeric freshness score** for every query using a learned estimator rather than a fixed decay rule.

FR5: Adaptive Retrieval Decision

The system shall decide whether **external retrieval is required** based on the freshness score and model confidence.

FR6: Retrieval Skipping

The system shall **skip retrieval** and answer directly from the LLM when its knowledge is estimated to be fresh, reducing latency and cost.

FR7: Evidence Retrieval

When retrieval is triggered, the system shall fetch relevant, up-to-date documents from **web search and/or a vector database**.

FR8: Evidence Ranking and Filtering

The system shall **rank and filter** retrieved documents by relevance and recency before passing them to the LLM.

FR9: Answer Generation

The system shall generate an answer using the LLM, **grounded in the retrieved evidence** whenever retrieval was performed.

FR10: Statement-Level Verification

The system shall split the generated answer into individual statements and **verify each statement against the retrieved evidence**.

FR11: Hallucination Detection

The system shall label each statement as **supported, unsupported, or contradicted** and flag potential hallucinations.

FR12: Corrective Action

The system shall **regenerate or correct** flagged statements, or warn the user, when unsupported content is detected.

FR13: Confidence Score Computation

The system shall compute a **confidence score (0–100%)** using model confidence, evidence support, and freshness awareness.

FR14: Evidence Highlighting

The system shall **highlight the supporting evidence** for each claim in the final answer to improve transparency and explainability.

FR15: Source Citation

The system shall display **source links, titles, and publication dates** for every piece of evidence used.

FR16: Final Answer Display

The system shall display the **final verified answer** together with its confidence score, freshness indicator, and evidence links.

FR17: Query History Storage

The system shall store **past queries, answers, scores, and sources** for user reference and system improvement.

FR18: User Feedback

The system shall allow users to **rate an answer or report an incorrect answer**, and store this feedback for later evaluation.

FR19: Admin Configuration

Administrators shall be able to configure **freshness and confidence thresholds, retrieval sources, and the LLM** used by the system.

FR20: Knowledge Base Management

Administrators shall be able to **add, update, or remove documents** in the vector database.

FR21: Performance Evaluation Module

The system shall evaluate and report **factuality, freshness accuracy, hallucination detection performance, and confidence calibration**.

FR22: Error Handling

The system shall display a clear message and **fall back gracefully** (e.g. answer with a low-confidence warning) if retrieval or the LLM fails.